Tetris-2.0

Team Sprint 1 Presentation (S1P)

Team: **Binary Ninjas**

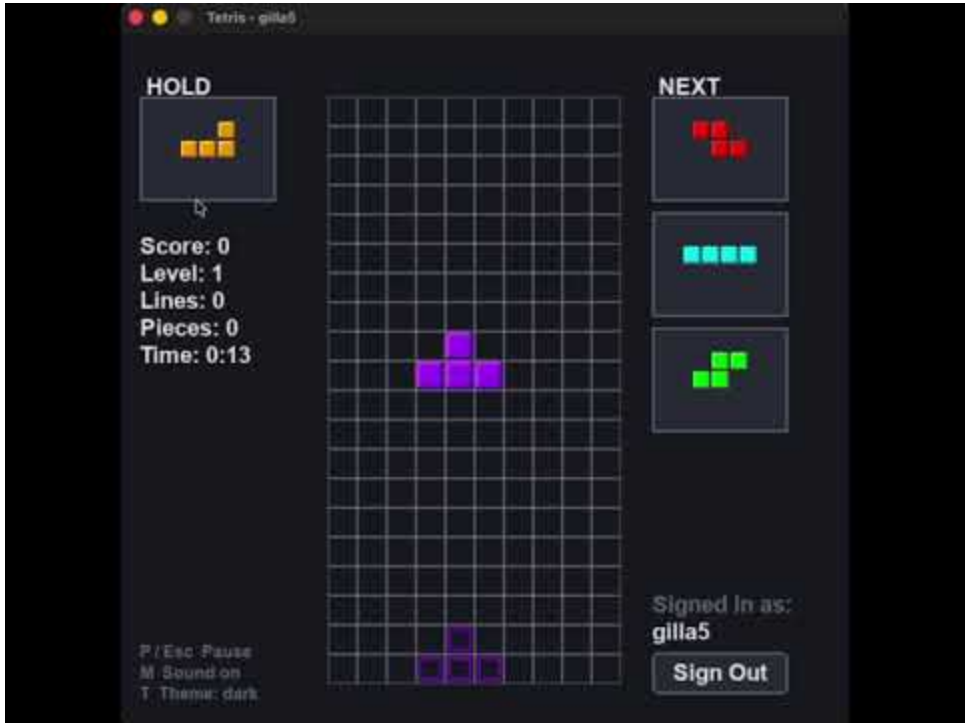
Team Leader: Aiden Gill

Members: Aiden · Nathan · Caleb · Jacob

Agenda

1. Sprint 1 demo (video)
2. Progress: lines of code and burndown
3. What went well, what went wrong, improvements
4. Sprint 2: goals, responsibilities, tests, timeline
5. Q&A

1. Sprint 1 Demo



Shows:

- Signing in with an account
- Playing a game: pieces, next queue, ghost piece
- Pause menu (Resume, Restart, Quit)

2. Progress: Lines of Code

Measure	Lines
Game code (`tetris_ver2.py`)	1,424
Test code (3 test files)	953
Team total	**2,377**

Member	Area	LoC
Aiden	Core engine and accounts	1,777
Nathan	Rendering and visual feedback	456
Caleb	Input and controls	144
Jacob	Audio and systems	0

2. Progress: Features and Requirements

Sprint 1 only (approved Sprint 1 scope):

- Features: 18 / 24 = 75%
- Requirements: 12 / 20 = 60%
- Remaining Sprint 1 requirements: 8

Whole project (counts only):

	Planned	Completed	Remaining
Features	24	18	6
Requirements	20	12	8

Automated tests: **109** (56 accounts, 10 rendering, 43 gameplay features)

2. Burndown: Remaining Requirements

[Insert burndown chart: remaining requirements vs. course week, with ideal line]

Week	Remaining
Start	14
End of Week 5	7
End of Week 6	0
End of Week 7	0
End of Week 8	0

3. What Went Well

- Accounts feature works end to end: sign up, sign in, per-player high scores
- 109 automated tests pass (accounts, rendering and gameplay features)
- Gameplay now includes pause, T-spins, combos, lock delay, sound, themes and animations
- Game code is organized into separate classes (`Board` , `Piece` , `SevenBag` , `Game`)
- Merge conflicts in the README and `tetris_ver2.py` were found and resolved

3. What Went Wrong

- Work was uneven: most code came from two members, so the input and audio tracks are behind
- Merge conflicts in `README.md` and `tetris_ver2.py` because changes lived on separate copies for too long
- Burndown is below plan, so the schedule needs adjusting
- Setup problems on Mac (pygame vs. pygame-ce on Python 3.14)

3. Cause and Improvements

Problem	Cause	Improvement
Uneven work	Tasks not broken into small pieces	Weekly check-in; small tasks with named owners
Merge conflicts	Long-lived changes, no branch routine	Small branches, pull before push, merge often
Burndown behind	Blocked or late tracks	Reassign work; update the schedule
Setup trouble	No standard environment	Adopt `uv` with a lockfile and `pygame-ce`

4. Sprint 2 Goals

- **Refactoring:** split `tetris_ver2.py` into a `tetris/` package (constants, pieces, game, render, accounts, scores, main), removing globals and magic numbers
- **MVP:** finish the must-have list: 7-bag, hold and next queue, SRS with wall kicks, soft and hard drop, line clears with speed, pause menu, game over, high scores
- **UV adoption:** `pyproject.toml`, `uv.lock`, `uv sync`, `uv run`
- **Quality:** README, setup guide and design document

4. Sprint 2 Responsibilities

Member	Responsibility
Aiden (Leader)	Engine refactor, `uv` migration, remaining engine feature, integration
Nathan	Rendering module, visual feedback, animations or themes
Caleb	Input module (key map, `handle_event`), pause menu, input tests
Jacob	Sound effects, music, mute toggle, audio tests

Proposed. Confirm each member before presenting.

4. Sprint 2 Test Plan

Type	Files
Unit	`test_pieces.py`, `test_board.py` (7-bag, rotation, wall kicks, collision, scoring)
Integration	`test_tetris_auth.py`, `test_game_loop.py` (accounts, spawn, drop, lock, clear, game over)
Regression	`test_regression.py` (same inputs before and after the refactor)
Acceptance	`test_acceptance.py` (one test per requirement) plus a manual play-through checklist

4. Sprint 2 Timeline

Weeks	Plan
9–10	Refactoring and core features; move to `uv`
11–12	Features, tests and design clean-up
13	Acceptance tests, documents, initial GitHub Pages site
14–15	Fixes and final submission

If burndown is not 100% by the end: adjust the schedule and submit a report.

Summary

- Sprint 1: accounts, engine, rendering and gameplay extras working; 109 tests; 2,377 lines
- Biggest risks: uneven workload and merge conflicts
- Sprint 2: refactor, `uv`, finish MVP, full test plan

Questions?